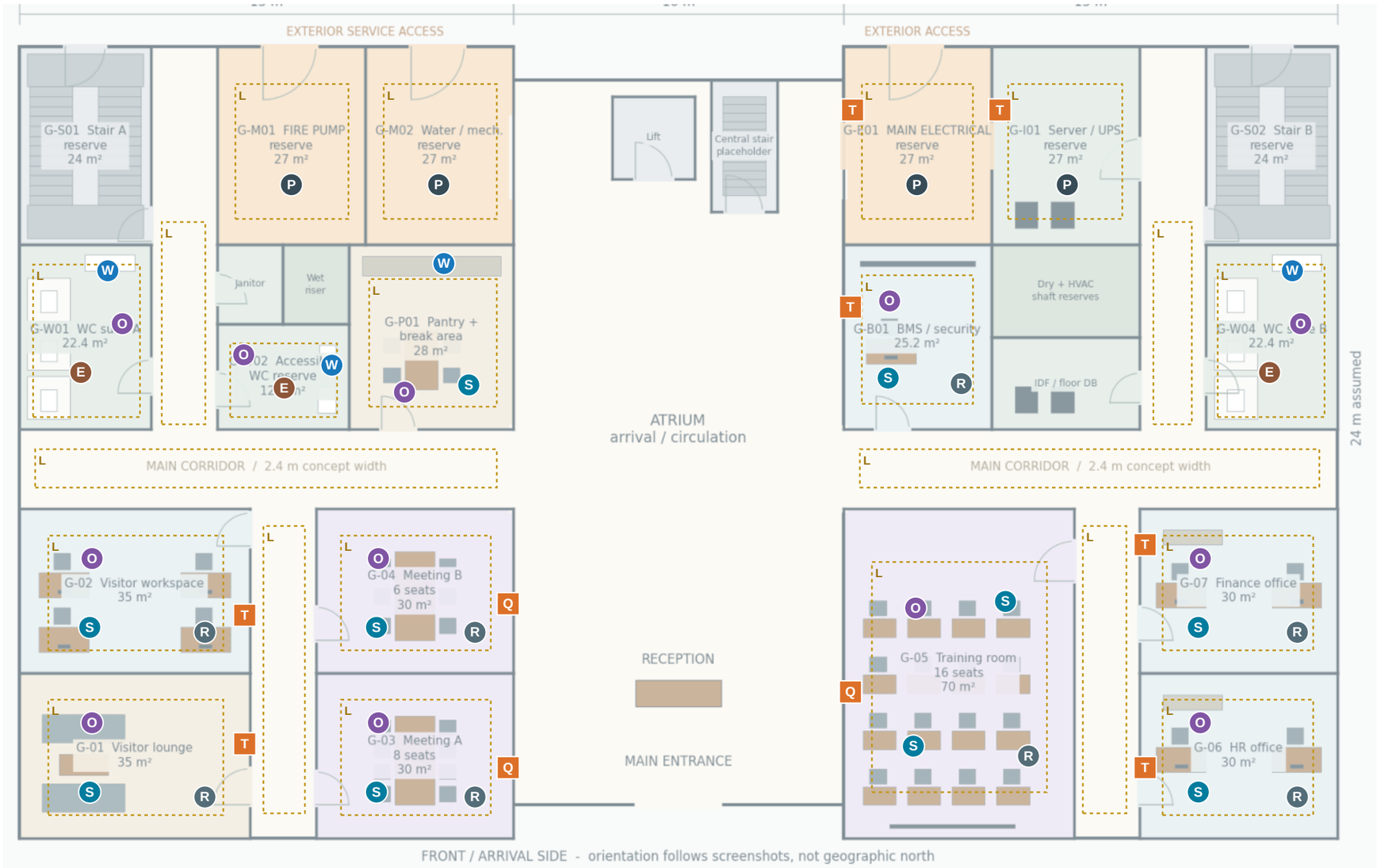
Ground floor | Proposed room-device locations

S/R/O/E symbols are ceiling zones; T/Q are wall candidates; W is low-level leak sensing. L outlines are lighting work zones, not individual luminaires.



FRONT / MAIN ARRIVAL SIDE

- **S — Supply-air outlet zone**
  Ceiling outlet(s) connect to the teal supply route. Airflow and outlet count remain unselected.
- **R — Return-air connection zone**
  Connect to the selected indoor-unit return or designed return/extract path. This is separate from supply.
- **O — Occupancy sensing candidate**
  Ceiling location to review against furniture, partitions and sensor coverage.
- **T — Temperature sensing candidate**
  Wall position near a representative internal-wall section; adjust for doors and local heat/drafts.
- **Q — Combined room sensing**
  Proposed temperature + CO2 sensing for meeting/training rooms; RH only if the chosen sensor provides it.
- **E — Toilet extraction zone**
  Ceiling grille(s) to the separate toilet-exhaust route; the fan may be located elsewhere.
- **W — Leak-sensing candidate**
  Under/near a credible leak source. Final location follows fixtures, equipment and floor drainage.
- **P — Plant instrumentation**
  See page 4: mount probes on the actual plant/pipe/tank, not at this room-tag position.

Ground-floor additions: reception/lounge lighting and sensing need coordination with the tall atrium. Pump, electrical and server-room P tags refer to page 4. **All floors:** place switches at accessible latch-side wall positions after checking each door swing. Existing exit signs/extinguishers are retained separately; their overlays are omitted here for clarity. Fire-device positions and stair emergency-light coverage remain design pending.

First floor | Proposed room-device locations

S/R/O/E symbols are ceiling zones; T/Q are wall candidates; W is low-level leak sensing. L outlines are lighting work zones, not individual luminaires.



FRONT / MAIN ARRIVAL SIDE

S Supply-air outlet zone
Ceiling outlet(s) connect to the teal supply route. Airflow and outlet count remain unselected.

R Return-air connection zone
Connect to the selected indoor-unit return or designed return/extract path. This is separate from supply.

O Occupancy sensing candidate
Ceiling location to review against furniture, partitions and sensor coverage.

T Temperature sensing candidate
Wall position near a representative internal-wall section; adjust for doors and local heat/drafts.

Q Combined room sensing
Proposed temperature + CO2 sensing for meeting/training rooms; RH only if the chosen sensor provides it.

E Toilet extraction zone
Ceiling grille(s) to the separate toilet-exhaust route; the fan may be located elsewhere.

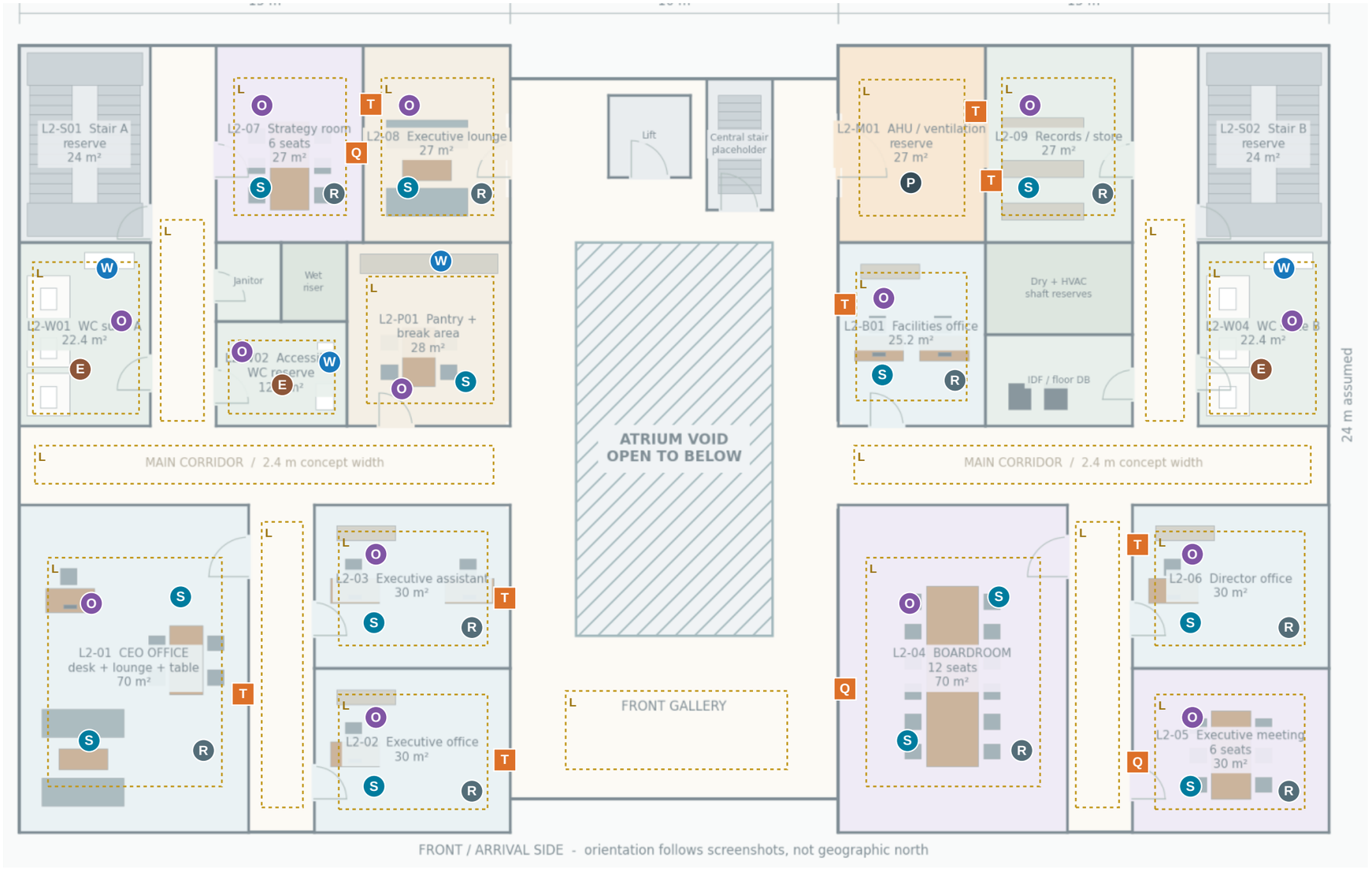
W Leak-sensing candidate
Under/near a credible leak source. Final location follows fixtures, equipment and floor drainage.

P Plant instrumentation
See page 4: mount probes on the actual plant/pipe/tank, not at this room-tag position.

First-floor focus: Q sensors are proposed in L1-07 Meeting D and L1-05 Meeting C. Team workspaces L1-01/L1-04 need coverage and HVAC zoning review; the illustrated O point is not proof that one sensor is sufficient. L1-08 print/archive needs a contents/equipment review. Ceiling branches connect to your service-route file; do not span the atrium void.

Second floor | Proposed room-device locations

S/R/O/E symbols are ceiling zones; T/Q are wall candidates; W is low-level leak sensing. L outlines are lighting work zones, not individual luminaires.



FRONT / MAIN ARRIVAL SIDE

- **S — Supply-air outlet zone**
  Ceiling outlet(s) connect to the teal supply route. Airflow and outlet count remain unselected.
- **R — Return-air connection zone**
  Connect to the selected indoor-unit return or designed return/extract path. This is separate from supply.
- **O — Occupancy sensing candidate**
  Ceiling location to review against furniture, partitions and sensor coverage.
- **T — Temperature sensing candidate**
  Wall position near a representative internal-wall section; adjust for doors and local heat/drafts.
- **Q — Combined room sensing**
  Proposed temperature + CO2 sensing for meeting/training rooms; RH only if the chosen sensor provides it.
- **E — Toilet extraction zone**
  Ceiling grille(s) to the separate toilet-exhaust route; the fan may be located elsewhere.
- **W — Leak-sensing candidate**
  Under/near a credible leak source. Final location follows fixtures, equipment and floor drainage.
- **P — Plant instrumentation**
  See page 4: mount probes on the actual plant/pipe/tank, not at this room-tag position.

Second-floor focus: Q sensors are proposed in L2-07 Strategy, L2-04 Boardroom and L2-05 Executive meeting. L2-01 CEO and L2-02/L2-03/L2-06 offices use the office package. L2-08 lounge and L2-09 records/store need their own use/contents review. The atrium remains a separate tall-volume detection, lighting and smoke-control study.

Finish the placement | Equipment, fire devices and connections

Use the markers as room-relative starting positions. They do not define final mounting heights, device quantities, penetration locations or code compliance.

LOCATION	WHAT TO PLACE AND WHERE	CONNECT IT TO
G-M01 fire pump room	Pressure transmitters at selected suction/discharge tappings; room temperature on a representative wall; leak sensing at credible leak points; lights above service aisles. Controller feedback is an electrical interface.	Fire-pump controller and approved monitoring interface. Keep running, available, fault and pressure separate.
G-M02 domestic water room	Pressure sensing on the booster pipework; leak probes beside water equipment; local lighting and controls. Tank-level sensor at the tank, not on a room wall.	Domestic controller/BMS inputs and local water routes.
G-I01 server/UPS and G-E01 electrical	Room temperature at a representative internal location; optional rack inlet sensing; suitable cooling terminals; leak sensing near actual condensate/water-risk paths. Keep equipment access clear.	Cooling controls, UPS interface and electrical meter/controller interfaces; data must exist in hardware.
L1-M01 and L2-M01 AHU rooms	Air-temperature probes in the intended ducts; filter differential-pressure taps across the filter; fan feedback at the controller. Lights and access panels must support maintenance.	AHU controller and BMS; link the actual measurement point to the correct AHU object.
Ordinary occupied rooms	Add fire detectors and sprinklers to the coordinated ceiling layer when their layout is established. Locate notification devices for the required audible/visible coverage. Do not infer one of each per room.	Detector/notification circuits to the FACP; sprinkler heads to fire-water branches. Ordinary sprinkler heads have no individual telemetry.
Corridors, galleries and stairs	Keep existing exit signs/extinguishers. Position emergency lights from the illumination design. Place required manual stations near the prescribed exits and notification appliances by coverage. Door contacts only on selected monitored doors.	Emergency lighting system; FACP circuits; door-contact inputs. These locations cannot be finalized from route centerlines alone.
Atrium	Reserve appropriate high-level detection, notification, sprinkler coverage and smoke-control equipment. Beam/aspirating detection are options, not selected devices. Account for height and maintenance access.	Approved fire/life-safety systems. Keep AURA monitoring-only for this prototype.

Copy into the 3D model: hide the roof; select a floor; place the S/R/O ceiling candidates and T/Q wall candidates; add lighting within the L work zones; place W probes at the actual fixture/equipment leak paths; connect each object to its service branch. Adjust all positions for walls, doors, beams, furniture, ceiling height and access.

Source and scope: the newly attached AURA_Main_Service_Routes(1).pdf repeats the first floor for four service networks. Ground/second-floor geometry here comes from the earlier Means_of_Egress_Fire_Extinguishers_BMS(2).pdf. These source drawings use an assumed 40 x 24 m envelope. No editable 3D model or measured reflected-ceiling plan was supplied, so exact model coordinates and fire coverage have not been established.

Model naming: use IDs such as Q-L1-07-01, O-L1-07-01 and S-L1-07-01, with room/floor/system fields. Store design status separately from data source (live/simulated/unconnected). A grille is a physical asset, not automatically a sensor; a command is not confirmed equipment operation.